Plant Physiology®

*Plant Physiology*, 2025, **198**, kiaf244
**https://doi.org/10.1093/plphys/kiaf244**
Advance access publication 6 June 2025
**Research Article**

# The peptide hormone RGF1 modulates PLETHORA2 stability via reactive oxygen species–dependent regulation of a cysteine residue

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## Abstract

The Root meristem growth factor 1 (RGF1) peptide extends the PLETHORA2 (PLT2) protein gradient by altering the distinct localization of superoxide ($O_2^-$) and hydrogen peroxide ($H_2O_2$) among the root developmental zones. However, the underlying mechanism through which reactive oxygen species (ROS) regulate PLT2 stability is unclear. Here, we demonstrate that the 212th cysteine of PLT2 is pivotal in modulating PLT2 stability through ROS. The PLT2 protein concentration gradient rapidly decreases in the elongation zone, where $H_2O_2$ accumulation initiates. However, substituting the 212th cysteine of PLT2 with serine ($PLT2^{C212S}$) results in PLT2 being more stable in the elongation zone, more broadly localized by RGF1, and showing robust resistance to $H_2O_2$. The sulfenylation of PLT2 was detected following treatment with $H_2O_2$ at high concentrations, suggesting that S-sulfenylation of the 212th cysteine controls PLT2 protein stability through local ROS distributions. These findings show that the formation of the PLT2 concentration gradient through ROS depends on a PLT2 sulfenylation mechanism that involves the 212th cysteine.

## Introduction

As sessile organisms, plants grow and develop organs in response to various environmental changes. Therefore, plants require real-time, sensitive cell-to-cell communications to control their development and adapt. Plant meristems enable plants to develop and produce new organs throughout their lifespan to optimize their growth. Plant roots show dynamic growth, which is tightly coordinated by each root zonation, i.e. the meristematic, elongation, and differentiation zones (Petricka et al. 2012). Stem cell daughters undergo active cell divisions in the meristematic zone to produce numerous small cells; after that, cells rapidly expand to increase their size and finally differentiate to acquire distinct structures (Petricka et al. 2012). The size of each developmental zone is determined by phytohormones, such as auxin and cytokinin, and master regulatory transcription factors—the *PLETHORA*s (*PLT*s)—which are critical for maintaining root stem cell activity and meristem growth (Perilli et al. 2012). *PLT* members contain 2 conserved *APETALA2/ETHYLENE-RESPONSE FACTOR* (*AP2/ERF*) DNA-binding domains (Aida et al. 2004; Galinha et al. 2007). The transcripts and promoter activities of the *PLT1*, *PLT2*, *PLT3*, and *BABY BOOM* (also known as *PLT4*) genes are detected around the quiescent center (Aida et al. 2004; Galinha et al. 2007). However, the proteins of PLTs display broader gradient distributions and partly overlap in the meristematic zone (Galinha et al. 2007), suggesting coordinated posttranslational regulation defines their gradients. In addition, they function redundantly in stem cell maintenance. PLT1 and PLT2 play particularly predominant roles in the root meristem (Aida et al. 2004; Galinha et al. 2007). PLTs dose-dependently control developmental events in the root developmental zones, suggesting the sophisticated PLT gradient defines root zonation (Galinha et al. 2007).

The sulfated ROOT MERISTEM GROWTH FACTOR (hereafter referred to as RGF, also known as GOLVEN or CLE-like) peptides also play pivotal roles in maintaining the root meristem and determining the root meristem size (Matsuzaki et al. 2010; Meng et al. 2012; Whitford et al. 2012). The *rgf* single mutant does not show obvious defects in the root, whereas the *rgf1/2/3* triple mutant reduces the root meristem size (Matsuzaki et al. 2010). External supply with sulfated synthetic RGF1 peptide rescues the *rgf1/2/3* root meristem defects, showing that the RGF1 dose-dependently functions in root meristem development (Matsuzaki et al. 2010). Our recent work showed that treatment with RGF1 expanded PLT2 protein localization by increasing superoxide ($O_2^-$) in the meristematic zone and decreasing hydrogen peroxide ($H_2O_2$) levels around the boundary between the meristematic and elongation zones (Yamada et al. 2020). The broadly localized PLT2 formed a larger root meristem (Yamada et al. 2020). Cotreatment with RGF1 and an NADPH oxidase inhibitor, which reduces $O_2^-$, or a high dose $H_2O_2$ (potent root growth inhibition), reduced the extended PLT2 localization (Yamada et al. 2020).

Received February 19, 2025. Accepted May 7, 2025.
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In contrast, cotreatment with RGF1 and an $H_2O_2$ scavenger, which diminishes $H_2O_2$, extended the PLT2 protein localization compared to the RGF1-only treatment (Yamada et al. 2020). These findings suggested that the increased $O_2^-$ and reduced $H_2O_2$ levels modulated by RGF1 extended the localization of the PLT2 protein. RGF1 treatment increased the PLT2 protein stability but not the *PLT2* promoter activity (Yamada et al. 2020). These findings suggested that the reactive oxygen species (ROS) distributions modulated by RGF1 posttranslationally alter the PLT2 protein stability. However, the underlying molecular mechanism through which the ROS affect the PLT2 protein stability in the root meristem remained obscure in these studies.

Here, we show that the 212th cysteine residue of the PLT2 protein is the most crucial component of ROS-dependent PLT2 stability. The PLT2 protein is sharply reduced in the elongation zone where $H_2O_2$ starts its accumulation and is susceptible to $H_2O_2$ treatment. However, replacing the 212th cysteine with serine ($PLT2^{C212S}$) of PLT2 resulted in the protein being stable even in the elongation zone. Furthermore, a broader gradient was observed, and there was a higher amount of $PLT2^{C212S}$ with a larger meristematic zone in comparison to replacement with another cysteine ($PLT2^{C112S}$) or the native form PLT2. Upon RGF1 treatment, the $PLT2^{C212S}$ protein localization and amounts were modulated to an even greater extent by ROS and showed potent resistance to $H_2O_2$ treatments. We detected more S-sulfenylation of PLT2 after $H_2O_2$ treatments than with mock treatment, suggesting that S-sulfenylation of the cysteine of the PLT2 protein via $H_2O_2$ modulates the PLT2 protein stability. Furthermore, the higher level of $PLT2^{C212S}$ protein modulated the expression of a similar array of gene expressions to native PLT2, but to a greater extent, suggesting that the $PLT2^{C212S}$ protein dose-dependently controls target-gene expressions by improving their protein stability through the 212th cysteine; however, it does not change the downstream genes. $PLT2^{C212S}$ upregulated the genes associated with cell proliferation expressed in the meristematic zone while simultaneously downregulating those genes linked to cell maturation localized in the elongation zone to a greater extent than the wild-type PLT2. Our results, thus, clearly demonstrate the posttranslational regulation and modification mechanisms of PLT2 protein stability via ROS upon RGF1 treatment, as well as the PLT2 downstream gene expressions under RGF1 signaling.

# Results

## PLT2 is a critical regulator in the RGF1 signaling pathway

The *plt1* and *plt2* single mutants form a root meristem of comparable size to those of the wild type (Galinha et al. 2007). However, the root meristem sizes of the *plt1/2* double mutants are significantly reduced (Galinha et al. 2007). The localization of these 2 PLT proteins overlaps in the apical root meristem (Galinha et al. 2007) (Fig. 1A), suggesting that the *plt1/2* mutants have a redundant function in root meristem development (Galinha et al. 2007). However, the PLT2 protein is more broadly distributed in the meristematic zone than the PLT1 protein (Galinha et al. 2007) (Fig. 1, A and C). We hypothesized that when the root meristem size was enlarged through RGF1 treatment, *PLT2* would be more involved in the RGF1 signaling due to the widespread localization of PLT2. To test our hypothesis, we examined the response of individual *plt* mutants to RGF1 peptide. We monitored the wild type, *plt1*, *plt2*, and *plt3* root phenotypes under a confocal microscope after 5 nM RGF1 treatment for 24 h. The meristematic zone size was dramatically increased in the wild type, *plt1*, and *plt3* after adding RGF1; however, the *plt2* mutant was almost insensitive to the peptide treatment with a meristematic zone size comparable to that of the mock treatment (Fig. 1, E and F), indicating that PLT2 is the dominant regulator elevating the meristematic zone size in the RGF1 signaling pathway.

We next investigated how RGF1 regulates the PLT1 and PLT 2 proteins by observing the protein localizations of gPLT1-yellow fluorescent protein (YFP) and gPLT2-YFP (Galinha et al. 2007) in the wild-type background upon supplying RGF1. The root meristem size was increased in both genetic backgrounds by RGF1, showing that RGF1 treatment works in both lines (Fig. 1, B and D). The gPLT2-YFP signal showed extended localization after adding RGF1 (Fig. 1, A and C). In contrast, the gPLT1-YFP signal was restricted around the root apical meristem irrespective of the RGF1 supplement (Fig. 1, A and C). In addition, we found that the gPLT1-YFP signal was slightly reduced after RGF1 treatment (Fig. 1, A and C), suggesting the distinct roles of PLT1 and PLT2 in regulating root meristem development under the RGF1 signaling pathway. This data again confirmed that PLT2 is the core component defining the meristem size in the RGF1 pathway. Furthermore, RNA-seq data for the genomic *PLT2* gene with a green fluorescent protein (*GFP*) introduced in the *plt2* mutant (the *gPLT2-GFP/plt2*) line showed that the *PLT1* expression levels were reduced after RGF1 treatment compared with the mock (Fig. 1G). However, in the *plt2* mutant, the *PLT1* expression was comparable even after the RGF1 treatment (Fig. 1G). The *PLT1* expression in the *gPLT2-GFP/plt2* line under the mock treatment was more reduced than that of the *plt2* mutant (Fig. 1G). These results demonstrated that the PLT2 transcriptionally reduced the *PLT1* expression under normal conditions and the RGF1 treatment. These results indicated that RGF1 transcriptionally decreased the *PLT1* expression via PLT2 but did not enhance the PLT1 protein allocation. These results show that PLT2 has a distinct function in the RGF1 signaling pathway.

## The Cys residue is essential for the modulation of PLT2 protein stability

Previously, we showed that ROS altered by RGF1 modulates PLT2 protein stability (Yamada et al. 2020). However, the underlying mechanism through which ROS control the PLT2 protein stability remained unknown. ROS are known to modify specific cysteine (Cys) residues to regulate protein stability and activity in various biological and physiological processes (Li et al. 2013; Paulsen and Carroll 2013). Two Cys residues appear among the PLT2 sequences, the 112th and 212th amino acids, and the latter residue is found in the *AP2/ERF* domain (Supplementary Fig. S1). Therefore, we hypothesized that the Cys residues of PLT2 are essential for the protein stability modulated by ROS. To test this hypothesis, we replaced 1 or both of the Cys residues of PLT2 with serine by site-directed mutagenesis. We replaced Cys with Ser because the difference between Cys and Ser is only the substitution of 1 sulfur atom with oxygen. The constructs of the single Cys substitution genomic $PLT2^{C112S}$ ($gPLT2^{C112S}$), $gPLT2^{C212S}$, and the double Cys substitution ($gPLT2^{C112SC212S}$), as well as the native *gPLT2* (hereafter designated as the wild-type *PLT2*) fused with green fluorescent protein (*GFP*) driven by the *PLT2* promoter, were introduced in the *plt2* mutant background for further analysis. We examined which construct rescues the RGF1 peptide-resistant phenotype of the *plt2* mutant since the *plt2* mutant showed strong resistance to RGF1 treatment (Fig. 1, E and F). Under the mock treatment, the meristem sizes of the $gPLT2^{C212S}$*-GFP/plt2* line were larger than those of the *gPLT2-GFP/plt2* and $gPLT2^{C112S}$*-GFP/plt2* (Fig. 2, A and B). All constructs recovered the RGF1 insensitive

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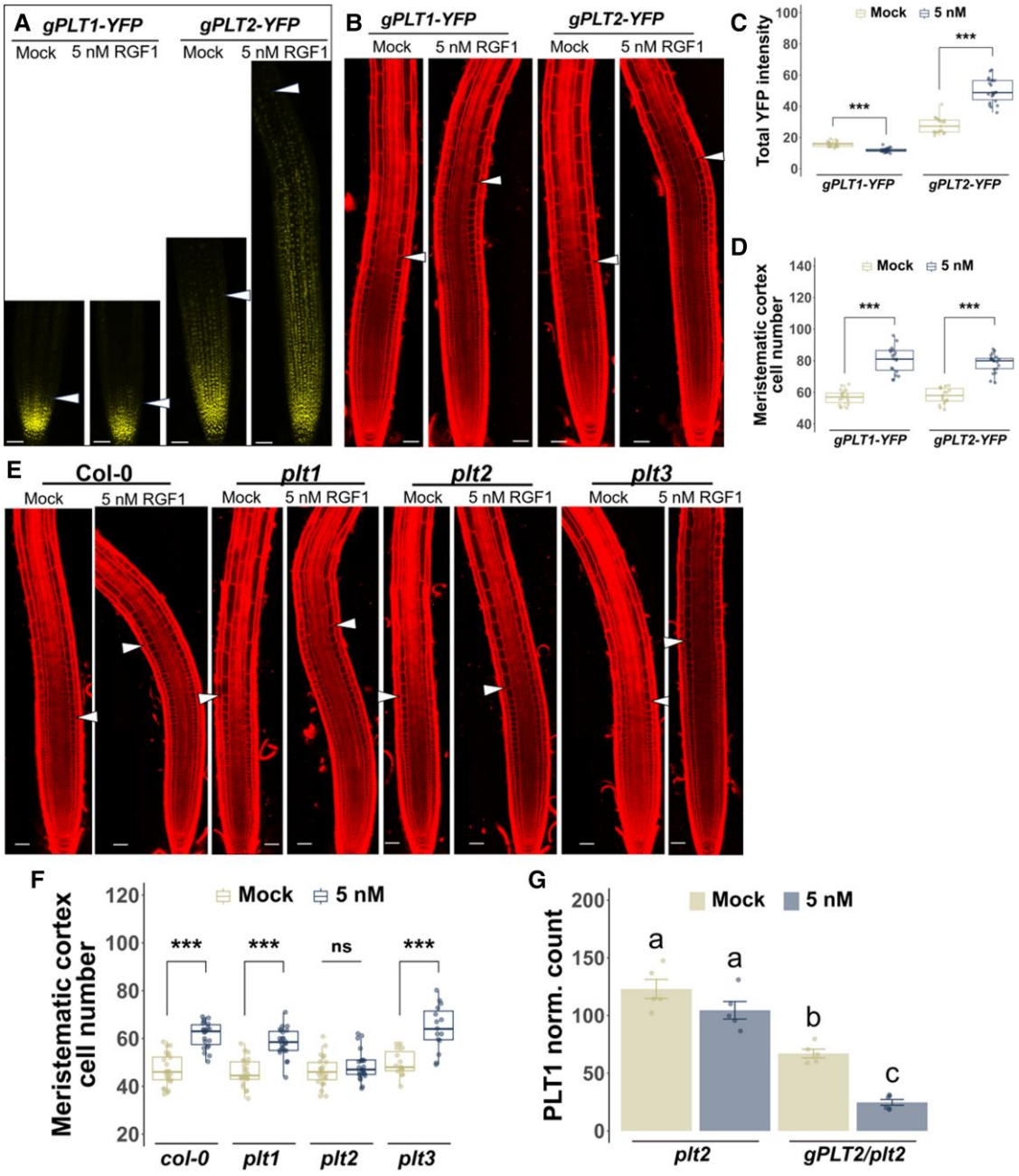



**Figure 1.** PLT2 protein is a critical regulator in the RGF1 signaling pathway. **A** to **D)** Confocal images **A** and **B)** and the quantitative analysis **C** and **D)** of *gPLT1-YFP/Col-0* and *gPLT2-YFP/Col-0* roots after mock or 5 nM RGF1 treatment for 24 h ($n = 20$; 3 biological replicates were performed, and a similar trend was observed). **E** and **F)** Confocal images **E)** and the meristematic cell numbers **F)** of *Col-0*, *plt1*, *plt2*, and *plt3* roots after mock or 5 nM RGF1 treatment for 24 h ($n = 24$; 3 biological replicates were performed, and a similar trend was observed). The image shown in **E)** is a composite image. **G)** The *PLT1* expression levels (normalized count) after mock or 5 nM RGF1 treatment for 24 h in designated genotypes from RNA-seq analysis. For **A)**, **B)**, and **E)**, the confocal microscopy image of the root sample is composed of multiple stitched tiles, set against a black background. For **B)** and **E)**, the propidium iodide (PI) staining is in red. The white arrowheads in **B)** and **E)** show the junction between the meristematic and elongation zones and in **A)** show the extent of YFP expression. All scale bars, 50 μm for **A)**, **B)**, and **E)**. In **C)**, **D)**, and **F)**, the group means were compared using a *t*-test (*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; nonsignificant (ns), $P > 0.05$). In the boxplots, the circles represent each individual data point; the horizontal line within boxes represents the median; the upper and lower hinges, respectively, indicate the 75th and 25th percentiles (IQR); and the whiskers show the 1.5 extension of the IQR. For **G)**, the pairwise comparisons were done using Tukey's HSD test (different letters indicate significant differences in mean, $\alpha = 0.05$). Error bars represent the standard error (SE) (the sample sizes are $n_{plt2\text{-Mock}} = 5$, $n_{plt2\text{-5nM}} = 5$, $n_{gPLT2/plt2\text{-Mock}} = 5$, and $n_{gPLT2/plt2\text{-5nM}} = 6$).

phenotypes of the *plt2* mutant. We could observe the root meristem enlarged phenotypes in all transgenic lines upon RGF1 (Fig. 2, A and B). Notably, the elevation of root meristem size in the 212th Cys substitution line was even more significant after RGF1 treatment than those of the wild type and the 112th Cys substitution lines (Fig. 2, A to C). However, the meristematic zone size of the *gPLT2*$^{C112S}$*-GFP* line was similarly increased by RGF1 compared to those of the wild-type PLT2 (Fig. 2, A to C). The 112th and 212th double Cys substitution line showed an equivalent potent response upon RGF1 treatment in comparison with the 212th single Cys substitution line, suggesting that the 112th Cys does not have an additive effect (Fig. 2, A to C). The 212th Cys substitution

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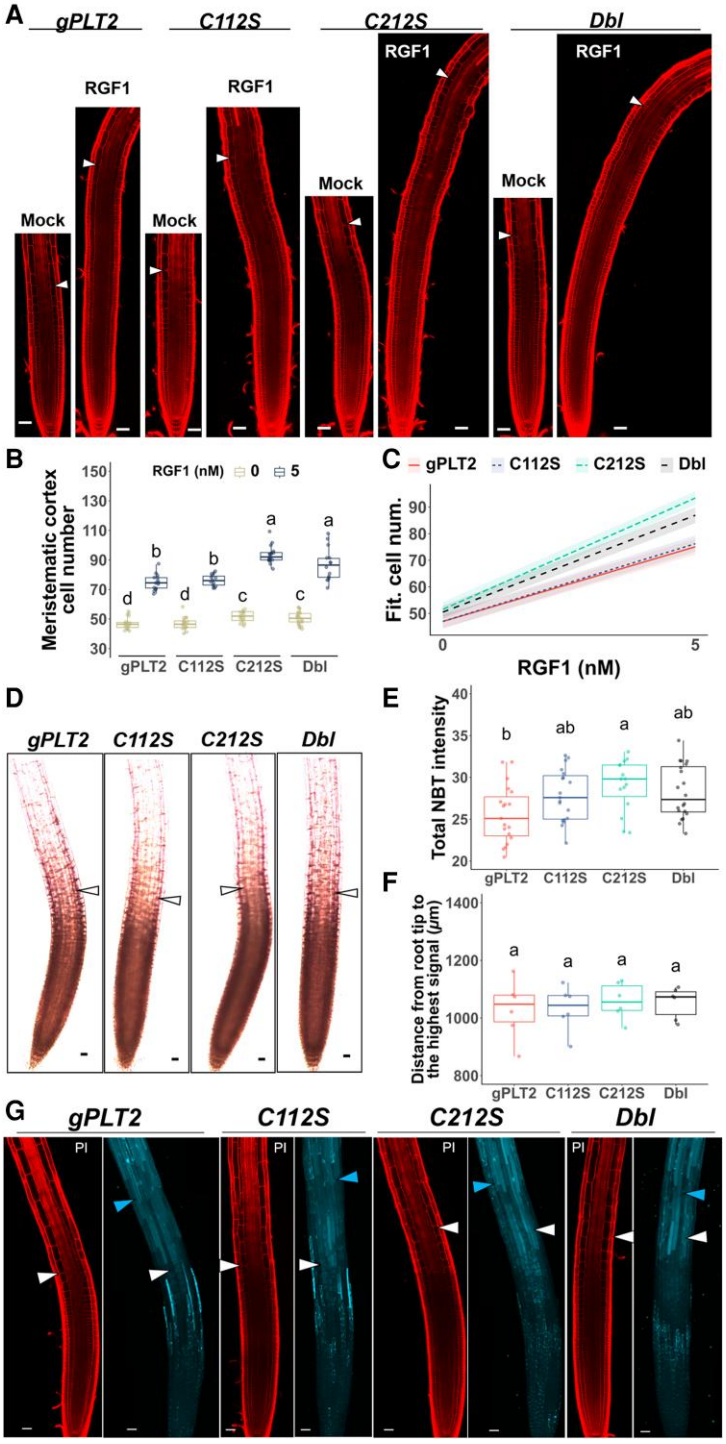



**Figure 2.** The ROS level was comparable in each PLT2 reporter line. **A)** Confocal images of the roots of designated genotypes after mock or RGF1 treatment for 24 h. **B)** Quantitative analysis of **A)** ($n = 20$; 3 biological replicates were performed, and a similar trend was observed). Letters represent significantly different means using the ART-ANOVA omnibus test followed by the ART-Contrast post hoc test with Benjamini–Yekutieli correction ($P < 0.05$). **C)** Interaction plots for **B)**, the response values were predicted after fitting with ordinal least square linear regression. The sample sizes are $n_{gPLT2} = 32$, $n_{C112S} = 31$, $n_{C212S} = 36$, and $n_{Dbl} = 36$. **D)** Light microscope images of the roots of the designated genotypes with NBT staining under normal conditions. **E** and **F)** Quantitative analysis of the roots of the designated genotypes under the indicated conditions; the pairwise comparisons were done using Tukey's HSD test ($n = 20$; 3 biological replicates were performed, and a similar trend was observed, and different letters indicate significant differences of mean, $\alpha = 0.05$). **E)** for **D)**; **F)** for **G)**. **G)** Confocal images of the roots of the designated genotypes with BES-$H_2O_2$-Ac staining under mock conditions. *gPLT2* indicates *gPLT2-GFP/plt2*; *C112S* indicates *gPLT2*$^{C112S}$*-GFP/plt2*; *C212S* indicates *gPLT2*$^{C212}$*-GFP/plt2*; and *Dbl* indicates *gPLT2*$^{C112SC212S}$*-GFP/plt2* genotypes. For **A)** and **G)**, the confocal microscopy image of the root sample is composed of multiple stitched tiles, set against a black background. The arrowheads in **A)** show the junction between the meristematic and elongation zones. For **B)**, **E)**, and **F)**, the circles represent each individual data point; the horizontal line within boxes represents the median; the upper and lower hinges, respectively, indicate the 75th and 25th percentiles (IQR); and the whiskers show the 1.5 extension of the IQR. The hollow arrowheads in **D)** indicate the junction between the meristematic and elongation zones. The hollow arrowheads in **G)** show the junction between the meristematic and elongation zones, and the top arrowheads in **G)** indicate the location of the highest BES-$H_2O_2$-Ac staining signal. All scale bar, 50 μm for **A)**, **D)**, and **G)**.

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lines formed larger meristematic zones with mock, and the effect was even more pronounced after RGF1 treatment (Fig. 2, A to C).

To confirm whether ROS levels are altered depending on RGF1 concentrations, we detected the $O_2^-$ and $H_2O_2$ levels in each PLT2 transgenic line by nitro blue tetrazolium (NBT) and $H_2O_2$-3′-O-acetyl-6′-O-pentafluorobenzenesulfonyl-2′-7′-difluorofluorescein-Ac ($H_2O_2$-BES-Ac) (Maeda et al. 2004; Yamada et al. 2020) staining after mock, 0.3 nM, and 5 nM RGF1 treatments. Under the mock treatment, we could observe the NBT signal in the meristematic zone (Fig. 2, D and E). Similarly, we could detect NBT signals in all PLT2 transgenic plants. The higher NBT staining in the meristematic zone was detected with 5 nM RGF1 treatment relative to mock and 0.3 nM RGF1 treatment in all genetic backgrounds (Supplementary Fig. S2, A and B). With the mock treatment, the $H_2O_2$-BES-Ac signals were strongly detected in the matured lateral root cap cells; however, they were not found in the other unmatured cells in the meristematic zone and were gradually increased in the elongation zone and highly accumulated in the differentiation zone (Fig. 2, F and G). Interestingly, the 0.3 and 5 nM RGF1 treatments strongly reduced the $H_2O_2$-BES-Ac signal at the boundary between the meristematic and elongation zones (Supplementary Fig. S2D). With 5 nM RGF1, the distance from the root tip to the highest $H_2O_2$-BES-Ac signal was further than those of the mock and 0.3 nM RGF1 treatment (Supplementary Fig. S2C), suggesting that RGF1 dose-dependently reduced the $H_2O_2$ signal over a broader area. These results confirmed that RGF1 dose-dependently modulated the 2 ROS similarly in all transgenic lines.

To understand how the different levels of the ROS affect the PLT2 stability via the Cys under the RGF1 signal, we examined the GFP signals after treatment with the 2 different concentrations of RGF1 in all transgenic lines. We found that the PLT2$^{C212S}$-GFP and PLT2$^{C112SC212S}$-GFP proteins displayed broader localization and higher intensities than the wild-type PLT2-GFP and PLT2$^{C112S}$-GFP upon each RGF1 treatment (Fig. 3, A to C). However, the localization of the PLT2$^{C112S}$-GFP was modestly increased by each application of RGF1 compared to the wild-type PLT2-GFP (Fig. 3, A to C). A similar trend was observed in the second independent transgenic line of each PLT2 construct (Supplementary Fig. S3, A to F). These results indicated that the 212th Cys residue is pivotal for the PLT2 stability through ROS in the RGF1 signaling pathway.

Furthermore, we conducted an immunoblot assay to examine the PLT2 protein amount after 0.3 nM RGF1 treatment by collecting the root tip tissues, including the 3 developmental zones from each PLT2 transgenic plant (Fig. 3E). Interestingly, the fold change of GFP protein amount relative to ACTIN (ACT) was elevated in the gPLT2$^{C212S}$-GFP and gPLT2$^{C112SC212S}$-GFP lines to a higher degree than the other 2 lines upon the RGF1 treatment (Fig. 3E, bottom of the graphs). The ratio of the GFP protein levels relative to ACT protein with RGF1 treatment in the gPLT2$^{C112S}$-GFP line was slightly increased more than that of the gPLT2-GFP line (Fig. 3E, bottom of the graphs). These results confirmed that the 212th Cys residue plays a vital role in regulating PLT2 protein stability through ROS by RGF1.

ROS levels were similarly modulated according to the concentration of the RGF1 treatment in the wild-type PLT2 and all the Cys substitution lines (Fig. 2, D to G; Supplementary Fig. S2). The 4 constructs had similar gene structures except for the sequences of 1 or 2 cysteines. These data thus suggested that RGF1 modulated PLT2 protein stability via the 212th Cys by ROS-dependent posttranslational regulation. To clarify this, we examined *PLT2* expression in each PLT2 genetic background using reverse transcription quantitative polymerase chain reaction (RT-qPCR). The *PLT2* expression relative to the *PROTEIN PHOSPHATASE 2A SUBUNIT A3* (*PP2AA3*) expression was comparable in all transgenic plants after the peptide treatment (Fig. 3D), suggesting that the broader localization of PLT2 protein in the 212th Cys substitution lines by RGF1 was caused by ROS-dependent posttranslational regulation but not transcriptional regulation.

We detected widespread PLT2 protein localization and larger meristematic zone size 24 h after RGF1 treatment (Figs. 2 and 3, A to C). It was still possible that the larger size meristem extends the PLT2 protein localization. To avoid the possibility that the larger meristematic zone increases the PLT2 protein distribution, we examined meristem sizes and GFP distributions in the wild-type PLT2 and the 212th Cys substitution lines by using a shorter time treatment (8 h RGF1 treatment). The RGF1 applications for 8 h slightly elevated the meristematic zone size in both lines compared to the mock treatment (Fig. 4, A to C). However, the meristem sizes with the short-time treatment were not enormously expanded compared with the 24 h RGF1 treatment (Figs. 2 and 4, A to C). Notably, the GFP signals in the 212th Cys replacement line were more strongly elevated and widespread than those in the wild-type PLT2, even after RGF1 treatment for the short period (Fig. 4, A and C). These results suggested that ROS alterations, rather than meristematic zone size, likely modulate the PLT2 protein stability.

Since it has been reported that the PLT2 protein moves from cell to cell (Mähönen et al. 2014), RGF1 may induce PLT2 protein movement. To ascertain whether the extended localization of PLT2 after RGF1 treatment was due to protein transfer, a construct with the wild-type *PLT2* fused to 3x*GFP* driven by the *PLT2* promoter was introduced in the *plt2* mutant background because the high molecular weight of the 3xGFP tag reduces the protein movement (Kurata et al. 2005). After RGF1 treatment, the PLT2-3xGFP protein was broadly localized at an equivalent level to the PLT2-1xGFP protein distribution (Supplementary Fig. S4, A and B), indicating that the broader localization of the PLT2-3xGFP protein was more likely to be enhanced by the PLT2 protein stability than the protein transport. Furthermore, immunoblot assay (Fig. 3E) showed that the protein amounts were highly elevated by RGF1 treatment in all PLT2 transgenic lines, including the Cys substitution lines, suggesting that RGF1 regulates the PLT2 protein stability via ROS rather than protein movement.

## ROS levels control PLT2 protein stability mainly via the 212th Cys

We quantified the GFP intensities of PLT2 without the RGF1 applications in the epidermal cells in each represented developmental zone, which were around the apical root meristem, the proximal side (close to the elongation zone) of the meristematic zone, and the elongation zone (Fig. 4D, red, yellow, blue arrowheads), in the wild-type *PLT2* and the *gPLT2$^{C212S}$-GFP* lines. The GFP signal in the wild-type PLT2 line decreased rapidly and became dim on the proximal side-meristematic (Fig. 4, D and E) and elongation zones, accumulating $H_2O_2$ (Fig. 2G). The PLT2$^{C212S}$ protein was more stable and broadly detected in these 2 developmental zones (Fig. 4D, yellow and blue arrowheads), thus showing a relatively strong signal in the cell expansion area with a higher chance of contact with $H_2O_2$ (Fig. 2G). Therefore, we hypothesized that $H_2O_2$ reduced the wild-type PLT2 protein level around the elongation zone; however, PLT2$^{C212S}$ was resistant to $H_2O_2$ and was still stable even at the cell elongation area. To test our hypothesis, we observed how the roots of each genetic background behaved by exogenously applying a high dose (1 mM, potent inhibition of root growth) of $H_2O_2$ for 24 h. The meristem size of the wild-type

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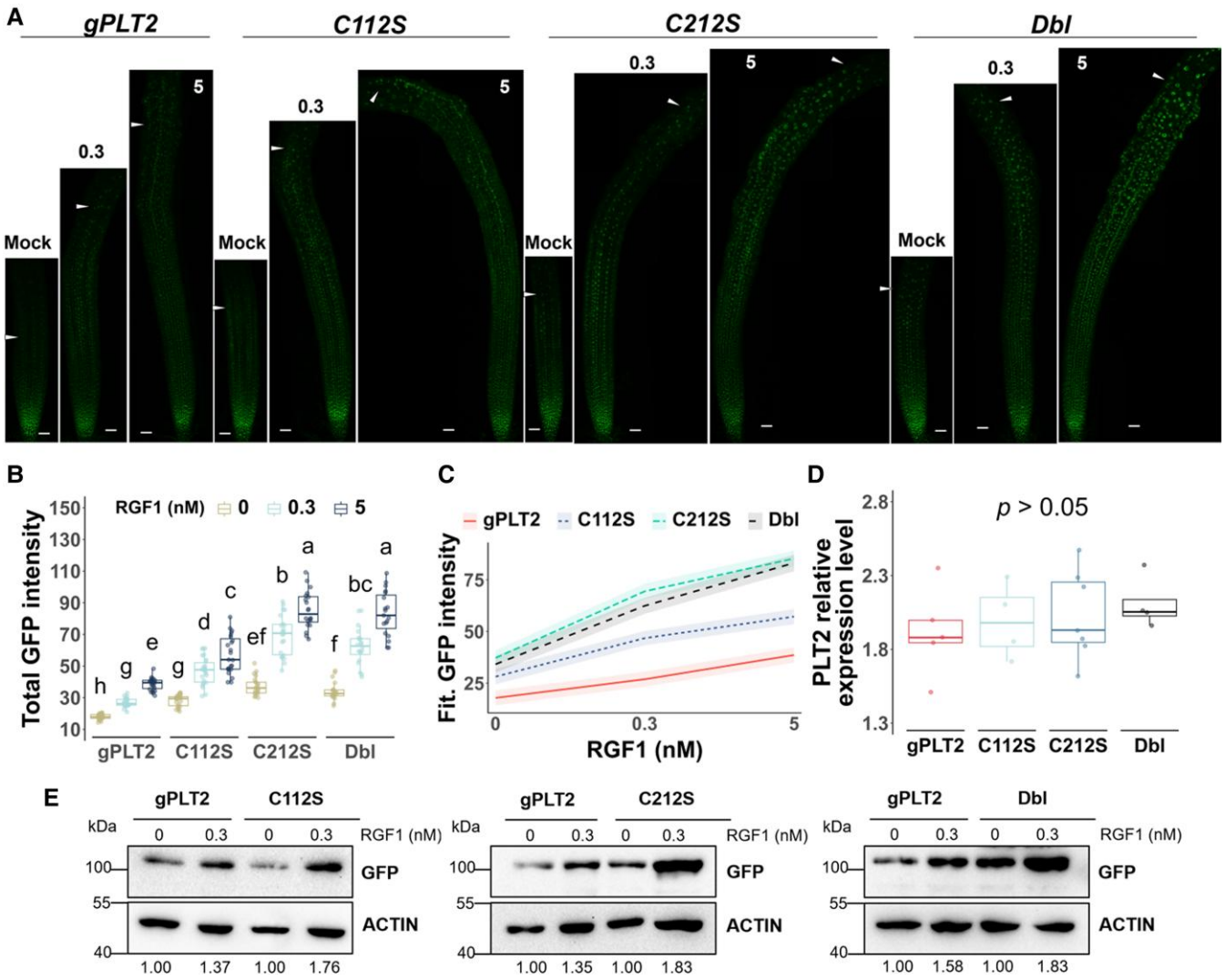



**Figure 3.** The cysteine residue of the PLT2 protein is essential for the RGF1-modulated PLT2 stability. **A)** Confocal images of the roots of designated genotypes after mock or RGF1 treatment for 24 h. The white arrowheads show the extent of GFP expression. The confocal image of the root sample is composed of multiple stitched tiles, set against a black background. All scale bar, 50 μm. **B)** Quantitative analysis of **A)** ($n = 20$; 3 biological replicates were performed, and a similar trend was observed). Letters represent significantly different means using the ART-ANOVA omnibus test followed by the ART-Contrast post hoc test with Benjamini–Yekutieli correction ($P < 0.05$). In the boxplots, the circles represent each individual data point; the horizontal line within boxes represents the median; the upper and lower hinges, respectively, indicate the 75th and 25th percentiles (IQR); and the whiskers show the 1.5 extension of the IQR. **C)** Interaction plots for **B)**; the response values were predicted after fitting with ordinal least square linear regression. The sample sizes are $n_{gPLT2\text{-}0} = 24$, $n_{gPLT2\text{-}0.3} = 23$, $n_{gPLT2\text{-}5} = 23$, $n_{C112S\text{-}0} = 22$, $n_{C112S\text{-}0.3} = 26$, $n_{C112S\text{-}5} = 23$, $n_{C212S\text{-}0} = 22$, $n_{C212S\text{-}0.3} = 22$, $n_{C212S\text{-}5} = 20$, $n_{Dbl\text{-}0} = 19$, $n_{Dbl\text{-}0.3} = 20$, and $n_{Dbl\text{-}5} = 19$. **D)** RT-qPCR quantification of the *PLT2* expression in the 3 developmental zones of the roots of the designated genotypes after either mock or 0.3 nM RGF1 treatment for 24 h, and no significant variation among means was observed from ANOVA ($n = 4$ to 5; each dot represents 1 biological replicate; $P > 0.05$). **E)** Western blot analysis of PLT2 protein in the roots of the designated genotypes after either mock or 0.3 nM RGF1 treatment for 24 h. ACTIN was the loading control. GFP normalized with ACTIN ratio is shown at the bottom of the graphs. *gPLT2* indicates *gPLT2-GFP/plt2*; *C112S* indicates $gPLT2^{C112S}$*-GFP/plt2*; *C212S* indicates $gPLT2^{C212}$*-GFP/plt2*; and *Dbl* indicates $gPLT2^{C112SC212S}$*-GFP/plt2* genotypes.

PLT2 was significantly reduced after high-concentration $H_2O_2$ treatment (Fig. 5, A and C). The wild-type PLT2-GFP and the $PLT2^{C112S}$-GFP signals were markedly decreased after $H_2O_2$ treatment (Fig. 5, B and D). Surprisingly, the $gPLT2^{C212S}$*-GFP* and $gPLT2^{C112SC212S}$*-GFP* lines demonstrated a robust resistance to $H_2O_2$, and the meristematic zone sizes treated with $H_2O_2$ were comparable and showed similar GFP intensities compared to the mock treatment (Fig. 5, A to D), suggesting that $H_2O_2$ mainly targets the 212th Cys to modulate PLT2 protein stability. Furthermore, we confirmed the PLT2 protein amounts by immunoblot analysis. The wild-type PLT2 and $PLT2^{C112S}$ proteins were labile when exposed to a high dose of $H_2O_2$, thus degrading sharply (Fig. 5E). However, the $PLT2^{C212S}$ and $PLT2^{C112SC212S}$ proteins were stable, even under high $H_2O_2$ conditions (Fig. 5E). These results thus revealed that the wild-type PLT2 protein was susceptible to $H_2O_2$, $PLT2^{C112S}$ showed a mildly resistant response, and $PLT2^{C212S}$ was almost insusceptible to $H_2O_2$ (Fig. 5E). We did not observe an additive effect in the PLT2 line with a double Cys replacement; therefore, we concluded that $H_2O_2$ was triggered to control the PLT2 protein stability predominantly on the 212th Cys.

Furthermore, to test the effect of $O_2^-$ on PLT2 protein stability, we treated RGF1 with a low concentration (700 nM, minor root growth defects) of diphenyleneiodonium (DPI) to reduce $O_2^-$ levels (Supplementary Fig. S5, A to C). A high dose of DPI, an NADPH oxidase inhibitor, strongly affects root development. However, 700 nM DPI mildly affected root meristem size in the wild-type

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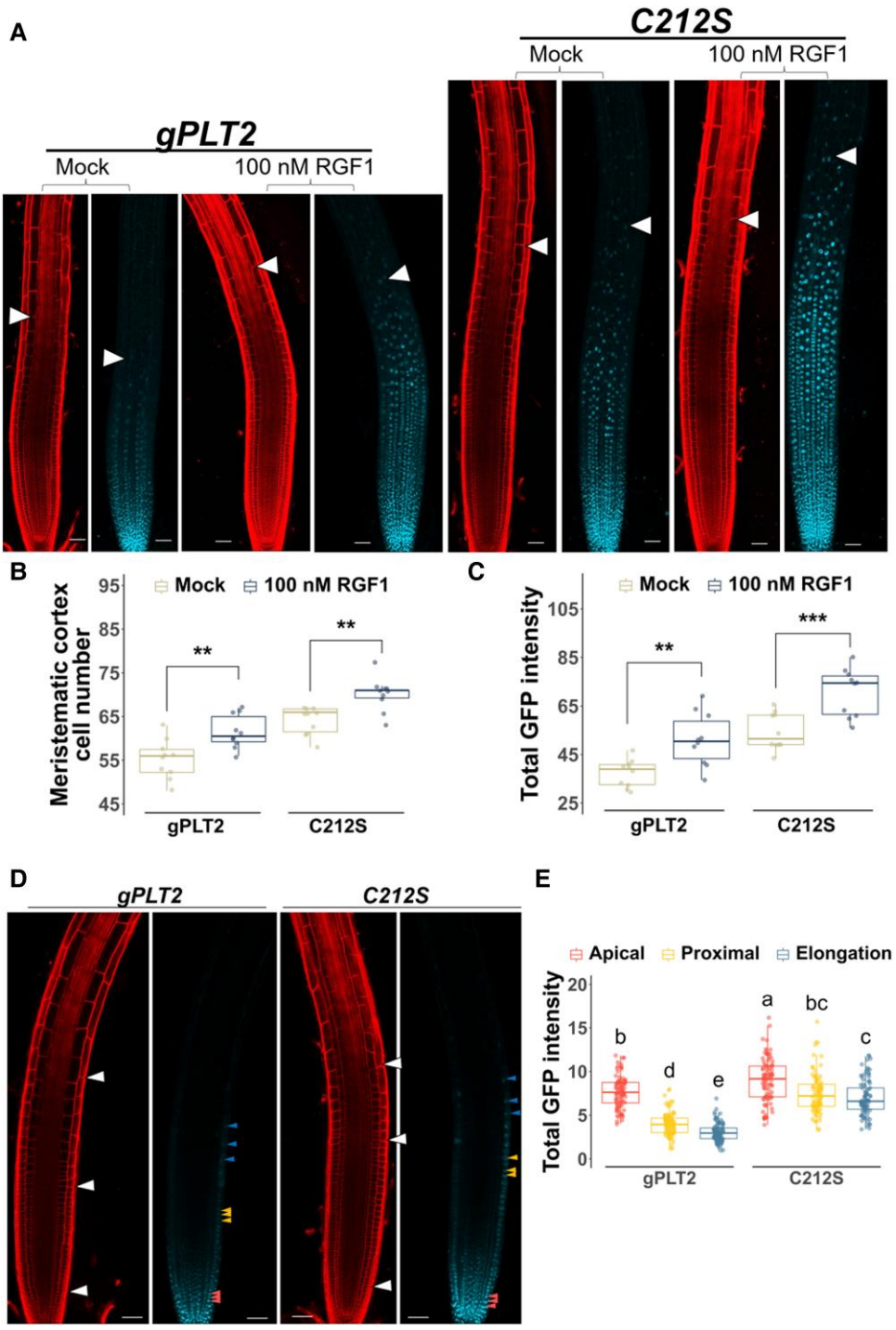



**Figure 4.** RGF1 controls the PLT2 protein stability via the 212th cysteine. **A)** Confocal images of the roots of the designated genotypes after either mock or 100 nM RGF1 treatment for 8 h. **B** and **C)** Quantitative analysis of the roots of the designated genotypes under the indicated treatments ($n = 10$; 3 biological replicates were performed, and a similar trend was observed). The group means were compared using a *t*-test (*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; nonsignificant (ns), $P > 0.05$). **D)** The PLT2 protein localization in the apical root meristem (bottom arrowheads), the proximal side of the meristematic zone (middle arrowheads), and the elongation zone (top arrowheads) in designated genotypes under normal conditions. The top, middle, and bottom white arrowheads in **D)** individually indicate the junction between the elongation and differentiation zone, the meristematic and elongation, and the apical meristem in the epidermal cells. **E)** Quantitative analysis of the roots of the designated genotypes under normal condition ($n = 10$; each dot represents the GFP intensity of single cell). The letters represent significantly different means using the ART-ANOVA omnibus test followed by the ART-Contrast post hoc test with Benjamini–Yekutieli correction ($P < 0.05$). In the **B)**, **C)**, and **E)** boxplots, the circles represent each individual data point; the horizontal line within boxes represents the median; the upper and lower hinges, respectively, indicate the 75th and 25th percentiles (IQR); and the whiskers show the 1.5 extension of the IQR. *gPLT2* indicates *gPLT2-GFP/plt2*; and C212S indicates *gPLT2*$^{C212}$*-GFP/plt2*. All scale bar, 50 μm for **A)** and **D)**.

PLT2 and Cys replacement lines (Supplementary Fig. S5, A to C). Cotreatment with RGF1 and DPI reduced PLT2 stabilities in the wild-type PLT2 and *gPLT2*$^{C112S}$*-GFP* lines markedly compared with RGF1 single treatment until they were a similar level to those of mock treatment (Supplementary Fig. S5, A to C). In contrast, the gPLT2$^{C212S}$ and gPLT2$^{C112SC212S}$ proteins were slightly reduced by

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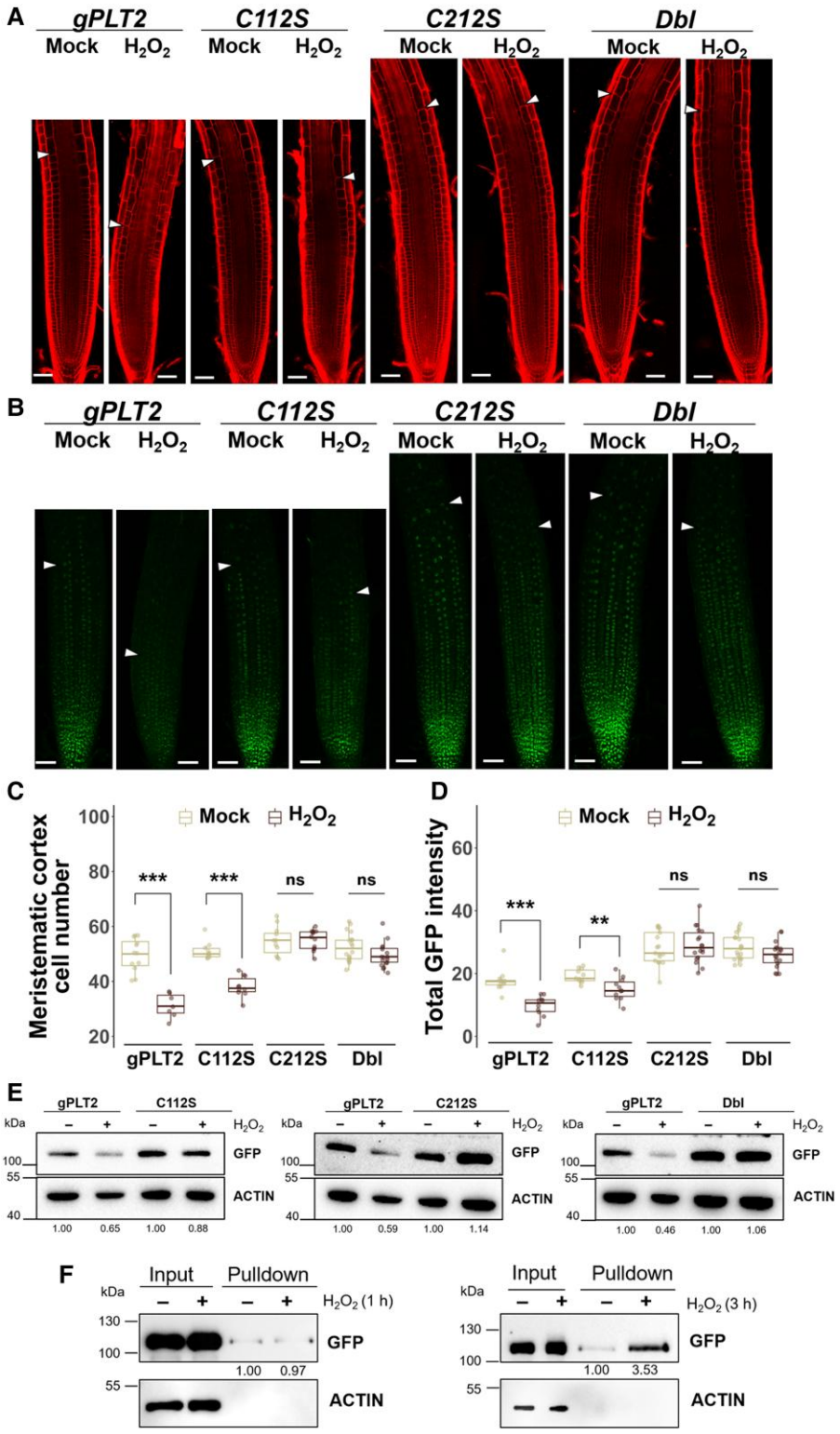



**Figure 5.** $H_2O_2$ controls the PLT2 protein stability via the 212th cysteine. **A** to **D)** Confocal images **A** and **B)** or quantitative analysis **C** and **D)** of the roots of designated genotypes after either mock or 1 mM $H_2O_2$ treatment for 24 h ($n = 15$; 3 biological replicates were performed, and a similar trend was observed). **E)** Western blot analysis of PLT2 protein in the roots of the designated genotypes after either mock or 1 mM $H_2O_2$ treatment for 24 h. ACTIN was the loading control. GFP normalized with ACTIN ratio is shown at the bottom of the graphs. **F)** Sulfenylation (SOH) of PLT2 in root tissues after either mock or 1 mM $H_2O_2$ treatment for 1 h (left panel) or 3 h (right panel). Sulfenylated PLT2 normalized with input ratio is shown at the bottom of the GFP graphs. *gPLT2* indicates *gPLT2-GFP/plt2*; *C112S* indicates $gPLT2^{C112S}$-*GFP/plt2*; *C212S* indicates $gPLT2^{C212}$-*GFP/plt2*; and *Dbl* indicates $gPLT2^{C112SC212S}$-*GFP/plt2* genotypes. The white arrowheads in **A)** and **B)** show the junction of the meristematic zone and the extent of GFP expression, respectively. All scale bar, 50 μm for **A)** and **B)**. For **C)** and **D)**, the group means were compared using a *t*-test (*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; nonsignificant (ns), $P > 0.05$). In the boxplots, the circles represent each individual data point; the horizontal line within boxes represents the median; the upper and lower hinges, respectively, indicate the 75th and 25th percentiles (IQR); and the whiskers show the 1.5 extension of the IQR.

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the cotreatment but not diminished to the degree of the mock treatment (Supplementary Fig. S5, A to C), suggesting that lower $O_2^-$ levels affect the PLT2 stability via the 212th Cys. The data from the $H_2O_2$ (Fig. 5, A to E) and DPI (Supplementary Fig. S5, A to C) treatments suggested that $O_2^-$ and $H_2O_2$ levels, respectively, stabilize and reduce the PLT2 protein through the 212th Cys.

## $H_2O_2$ induces S-sulfenylation of the PLT2 protein

The PLT2 proteins are localized in the meristematic zone, where $O_2^-$ accumulates highly (Figs. 2D). The wild-type PLT2 started to be degraded in the elongation zone (Fig. 4, D and E), where $H_2O_2$ began accumulating (Fig. 2G). The wild-type PLT2 was susceptible to $H_2O_2$ treatment (Fig. 5, B, D, and E). However, the PLT2$^{C212S}$ protein was still localized in the elongation zone and was insensitive to the $H_2O_2$ treatment (Fig. 5, B, D, and E), suggesting that $H_2O_2$ treatment reduces the PLT2 protein stability by modification of the 212th Cys of the PLT2 protein. $H_2O_2$ induces the posttranslational modification of Cys from thiol (-SH) to sulfenic acid (-SOH), named sulfenylation, in various organisms (Huang et al. 2019). Recent research has shown that $H_2O_2$ induces S-sulfenylation of the Cys of target proteins to modulate the protein functions in *Arabidopsis* (Liu et al. 2022; Cao et al. 2024). We hypothesized that $H_2O_2$ treatment involves sulfenylation of PLT2 protein. Since PLT2 protein accumulates only in the meristematic zone, extracting high amounts of the PLT2 protein to detect specific modifications is technically tricky. Therefore, we inducibly overexpressed the wild-type PLT2-GFP in a whole root for 24 h using an estradiol-inducible system (Zuo et al. 2000; Curtis and Grossniklaus 2003). We applied mock and 1 mM $H_2O_2$ treatment for 1 and 3 h after the 24 h inducible overexpression of *PLT2* to examine the PLT2 protein modification. We detected more S-sulfenylation using a biotin-switch method (Cao et al. 2024) than the mock treatment at 3 h after the $H_2O_2$ treatment compared with the 1 h treatment (Fig. 5F). These results strongly suggested that $H_2O_2$ modified sulfenylation of the 212th Cys of the PLT2 protein to modulate the stability.

## PLT2$^{C212S}$ modulates specific root development-related downstream genes after RGF1 treatment to a higher extent than the wild-type PLT2

High levels of PLT control stem cell maintenance, moderate levels promote cell proliferation, and lower levels are required for cell differentiation, suggesting the sophisticated PLT gradient defines root development and zonation (Galinha et al. 2007; Mähönen et al. 2014). Biochemistry experiments and microscopic analysis showed that the 212th Cys replacement decelerated the PLT2 degradation, enlarged the meristematic zone size, and delayed the initiation of the cell elongation and differentiation with the mock treatment to a greater extent than the wild-type PLT2 line (Fig. 2, A to C). These phenotypes and the PLT2$^{C212S}$ protein amount were enhanced by the RGF1 treatment in the PLT2$^{C212S}$ line to a greater extent than the wild-type PLT2 (Fig. 3, A to C). Therefore, we hypothesized that a higher amount of PLT2$^{C212S}$ affected more downstream genes required for the specific root developmental zones. To assess the impacts of PLT2 with or without the 212th Cys in the *plt2* mutant at the transcriptome level, we performed RNA sequencing (RNA-seq) analysis in root tips containing the 3 developmental zones 24 h after mock, 0.3, and 5 nM RGF1 treatment. Differentially expressed gene (DEG) analysis was conducted for the *plt2* mutant, *gPLT2-GFP/plt2*, and *gPLT2*$^{C212S}$*-GFP/plt2* between the mock and RGF1 treatments (Fig. 6, A to B; Supplementary Data S1). The *gPLT2-GFP/plt2* line had over 8,500 DEGs (4,083 upregulated genes [Fig. 6A, yellow circle] and 4,477 downregulated genes [Fig. 6A, green circle]). However, the *gPLT2*$^{C212S}$*-GFP/plt2* had over 10,500 DEGs (up- and downregulated genes, 5,275 [Fig. 6A, red circle] and 5,317 [Fig. 6A, gray circle], respectively). These results suggested that the PLT2$^{C212S}$ protein modulated the downstream genes more than the wild-type PLT2 protein. We compared the up- and downregulated DEGs in the *gPLT2-GFP* and *gPLT2*$^{C212S}$*-GFP* lines (Fig. 6B). Three thousand and four hundred sixty-four positively and 3,857 negatively regulated DEGs overlapped (Fig. 6B, inside the intersection in the Venn diagram). Over 84% (3,464/4,083) of upregulated DEGs and 86% (3,857/4,477) of downregulated DEGs in the *gPLT2-GFP/plt2* line overlapped with those in the *gPLT2*$^{C212S}$*-GFP/plt2* line (Fig. 6B); moreover, the common DEG genes among them were more profoundly modulated in *gPLT2*$^{C212S}$*-GFP* than in *gPLT2-GFP/plt2* (Fig. 6C). These results suggested that the PLT2$^{C212S}$ protein modulates expressions of similar target genes in a dose-dependent manner to a greater extent than the wild-type PLT2 protein.

The DEG data were further compared with our previous root developmental zone-specific transcriptome analysis (Li et al. 2016) to understand which developmental zone-specific expression genes the PLT2 and PLT2$^{C212S}$ proteins regulate at the transcript level. We found that 1,098 and 377 genes were expressed individually in the meristematic zone (Fig. 6D) and the elongation zone (Fig. 6E) to a more significantly potent extent than in the other 2 developmental zones. Over 80% (883/1,098) of gene expressions of the meristematic zone-specific genes (1,098 genes) were more upregulated in the PLT2$^{C212S}$ line with each RGF1 treatment compared to in the PLT2 wild-type line (Fig. 6D). In contrast, over 89% (339/377) of the gene expressions of the elongation zone-specific genes (377 genes) were more downregulated in the 212th Cys substitution line than in the PLT2 wild-type line (Fig. 6E). These results demonstrated that the PLT2$^{C212S}$ protein modulated the developmental zone-specific genes more potently due to the higher PLT2$^{C212S}$ protein amount. Furthermore, to examine which developmental event PLT2 and PLT2$^{C212S}$ proteins regulate, we conducted a Gene Ontology (GO) analysis using common positively and negatively regulated DEGs in each developmental zone (Fig. 6, F and G). The GO highly enriched categories of the 883 upregulated genes were related to cell proliferation (Fig. 6F; Supplementary Data S2).

On the other hand, GOs relevant to cell differentiation and cell wall biogenesis were significantly enriched in the 339 downregulated genes (Fig. 6G; Supplementary Data S2). In addition, we used RT-qPCR to validate the upregulation of genes related to the mitotic cell cycle in the PLT2$^{C212S}$ line (Fig. 6H) and the downregulation of genes involved in cell wall biogenesis in the PLT2$^{C212S}$ line after 0.3 nM RGF1 supplement (Fig. 6I). These data confirmed that the highly accumulated PLT2$^{C212S}$ protein controlled morphological change to a greater extent than the PLT2 protein by modulating the expression of developmentally specific genes.

# Discussion

ROS distributions altered by RGF1 affect the developmental transitions in the root meristem by enhancing PLT2 stability (Yamada et al. 2020). However, how the discrete ROS in each developmental zone affected the PLT2 protein stability was unclear. In this study, we provided evidence that ROS modulate the stability of PLT2 by influencing its Cys residues, especially the 212th (Figs. 2 to 5). The *PLT2* mRNAs and *PLT2* promoter activity were detected explicitly around the RAM (Aida et al. 2004; Galinha et al. 2007). However, the gPLT2-GFP protein signal formed a concentration gradient from the RAM to the elongation zone (Fig. 3A), suggesting

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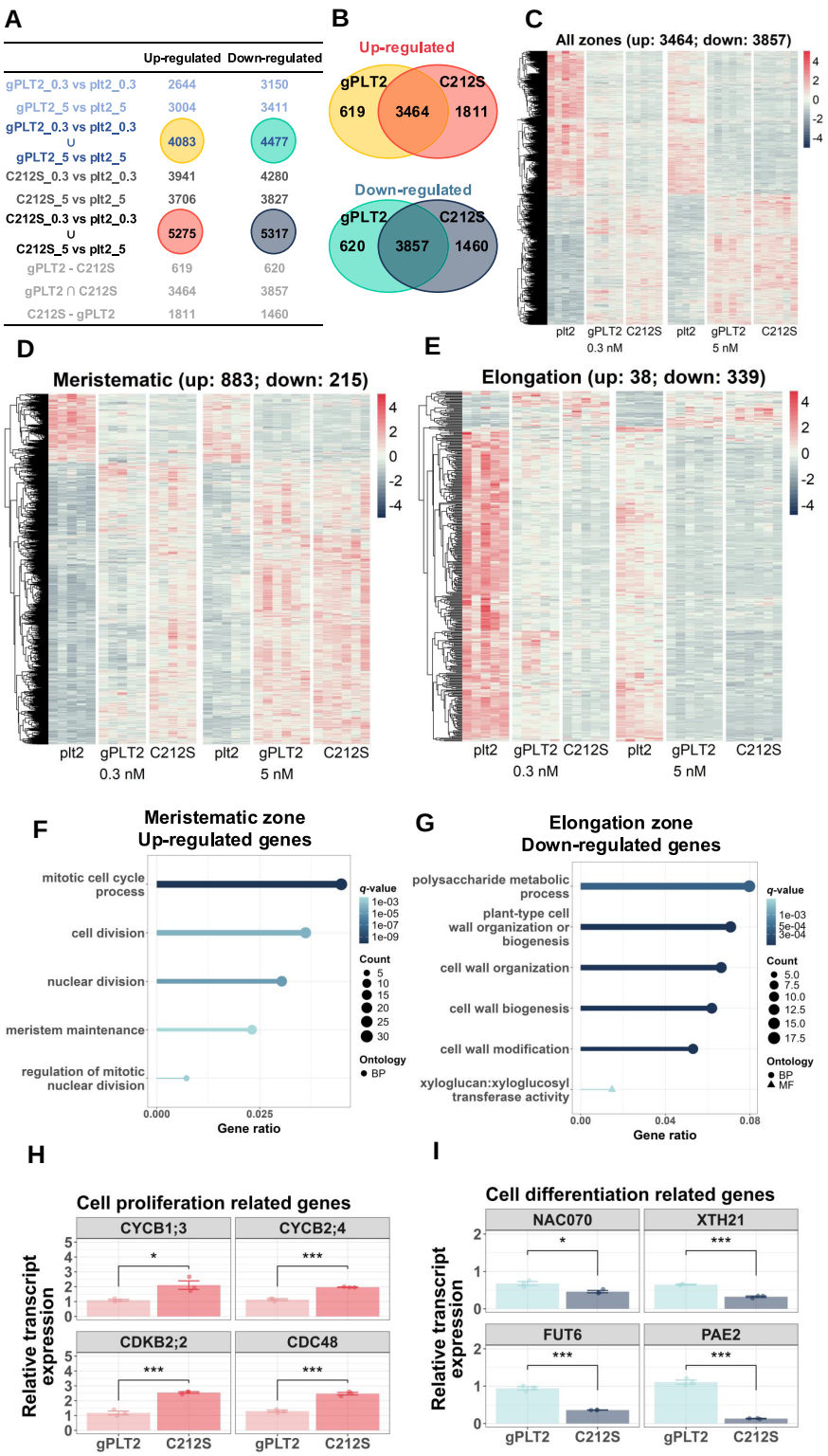



**Figure 6.** PLT2$^{C212S}$ induces similar genes to PLT2 after RGF1 treatment but to a greater extent. **A)** The number of upregulated and downregulated DEGs (FDR < 0.05) in the *gPLT2* and *C212S* lines compared to the *plt2* upon different concentrations of RGF1 treatment. **B)** The Venn diagram shows the overlap of the upregulated and downregulated DEGs in the *gPLT2* and *C212S* lines. The intersection between the *gPLT2* and *C212S* was extracted to conduct subsequent analysis. **C)** A heatmap of the upregulated and downregulated gene sets (compared to *plt2*) from the intersected gene sets in **B)**. **D** and **E)** Genes from **C)** which are expressed in the meristematic zone **D)** or elongation zone **E)**. The zone-specific expressing genes were defined as the expression ratio in the meristematic zone **D)** or elongation zone **E)** being more than 0.65 based on the data in (Li et al. 2016). **F)** Cell proliferation-related GO terms yielded from the upregulated genes in **D)**. **G)** Cell differentiation-related GO terms yielded from the downregulated genes in **E)**. **H** and **I)** RT-qPCR validation of the genes related to the mitotic cell cycle **H)** and cell wall biogenesis **I)** in the gPLT2 and C212S lines after 0.3 nM RGF1 treatment. Error bars represent the standard error (SE) ($n = 3$; each dot represents 1 biological replicate). The group means were compared using a *t*-test (*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$; nonsignificant (ns), $P > 0.05$). *gPLT2* indicates *gPLT2-GFP/plt2*; and *C212S* indicates *gPLT2*$^{C212}$*-GFP/plt2*. For **C)**, **D)**, and **E)**, the color scale bar represents the row-wise centered and scaled values for the normalized count of the gene expression.

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that the PLT2 protein was synthesized from mRNAs around the RAM and gradually broken down all the way up to the elongation zone, where $H_2O_2$ starts accumulating (Dunand et al. 2007; Yamada et al. 2020) (Fig. 2G). These results suggested that the PLT2 protein was slowly degraded in the meristematic zone (Fig. 3A), where $O_2^-$ is highly produced (Fig. 2D). The PLT2 protein was rapidly reduced in the elongation zone (Fig. 4, D and E), where $H_2O_2$ initiates its accumulation (Fig. 2G). In fact, $H_2O_2$ treatment strongly decreased the wild-type PLT2 protein amount (Fig. 5, B and E). However, the PLT2$^{C212S}$ protein remained more stable in the elongation zone (Fig. 4, D and E) and had intact protein amounts even after the $H_2O_2$ treatment (Fig. 5, B, D, and E). Finally, we demonstrated that $H_2O_2$ induced sulfenylation of the PLT2 protein (Fig. 5F). These results suggest that $O_2^-$ maintains the free thiol (S-H) of the 212th Cys in the meristematic zone and $H_2O_2$ sulfenylates the Cys (S-OH) in the elongation and differentiation zones (Supplementary Fig. S6). The sulfenylated PLT2 is presumably sulfinylated (S-$O_2$H) and sulfonated (S-$O_3$H) by continuous oxidization of $H_2O_2$ in the elongation and differentiation zone and finally degraded. However, RGF1 treatment strongly reduced $H_2O_2$ levels at the boundary between the meristematic and elongation zones and increased $O_2^-$ concentrations in the meristematic zone (Yamada et al. 2020) (Supplementary Figs. S2 and S6). The reduced $H_2O_2$ and increased $O_2^-$ by the RGF1 treatment reversibly produced free thiol from sulfenylated PLT2 protein. These findings suggest that the 212th Cys is substantially affected by local ROS alteration at specific root sites, presumably by reversible modification from S-sulfenylation to free thiol to expand the concentration gradients after the RGF1 treatment.

The PLT2$^{C212S}$ protein amount was highly increased by the RGF1 treatment (Fig. 3E; please see the fold change below). The PLT2$^{C112S}$ protein amount and the PLT2$^{C112S}$-GFP signal were still slightly increased compared with the wild-type PLT2 by RGF1 (Fig. 3, A, B, and E). However, these alterations were lower than the PLT2$^{C212S}$ protein (Fig. 3E; please see the fold change below; Fig. 3, A and B). Furthermore, PLT2$^{C112SC212S}$ protein amounts were similarly increased to the PLT2$^{C212S}$ protein, suggesting that the 112th cysteine did not have an additive effect. The PLT2 protein dose-dependently controls the root meristem size. The RGF1 treatment modulated the PLT2$^{C112S}$ protein amount at molecular levels at a slightly higher extent than the wild-type PLT2. However, the meristematic zone size of the gPLT2$^{C112S}$-GFP lines was comparable to the wild-type PLT2 line even after RGF1 treatment (Fig. 2B). The alteration of the PLT2$^{C112S}$ protein amount upon RGF1 was not enough to extend the meristematic zone size (Fig. 2, A and B), suggesting that the 112th cysteine does not primarily contribute to the meristematic zone size control.

Furthermore, the PLT2$^{C212S}$ protein was highly resistant to $H_2O_2$. In contrast, the PLT2$^{C112S}$ protein was susceptible to $H_2O_2$. However, the gPLT2$^{C112S}$-GFP line had slightly higher GFP signals and PLT2$^{C112S}$ protein amounts than the wild-type PLT2 (Fig. 5, D and E). The PLT2$^{C112SC212S}$ did not have an additive effect. The stable PLT2$^{C212S}$ protein resulted in resistance to $H_2O_2$. The PLT2 protein dose-dependently controls the root meristem size (Galinha et al. 2007). The reduced PLT2$^{C112S}$ protein levels upon $H_2O_2$ diminished the meristematic zone size to a similar extent to the wild-type PLT2 (Fig. 5C), suggesting that the reduction levels of PLT2$^{C112S}$ protein upon the $H_2O_2$ treatment contributed to shrunk meristematic zone size.

We could detect moderate alterations of the PLT2$^{C112S}$ protein amount at molecular levels after the RGF1 and $H_2O_2$ treatments. However, these alterations of the PLT2$^{C112S}$ protein amount did not greatly result in the control of expansion of the meristematic zone size and the resistance to $H_2O_2$ compared with the PLT2$^{C212S}$ protein. The alterations of the PLT2$^{C112S}$ protein were detectable at the molecular levels. However, these were not enough for the alteration of the phenotype. These suggested that the 212th cysteine has a primary role in controlling the size of the meristematic zone.

The 212th Cys is conserved in the DNA-binding domain of PLT1 and PLT2 (Supplementary Fig. S1). Even so, only PLT2 showed a significant effect in response to RGF1 via ROS (Fig. 1, A to F). RGF1 significantly decreases $H_2O_2$ levels around the boundary between the meristematic and elongation zones (Supplementary Fig. S2, C and D) (Galinha et al. 2007). The PLT2 concentration gradient is formed until the region where RGF1 affects the $H_2O_2$ distributions (Figs. 1A and 4, D and E). The $H_2O_2$ levels initiated to increase around the boundary (Fig. 2G). The wild-type PLT2 protein was rapidly degraded before the boundary (Fig. 4, D and E). However, the PLT2$^{C212S}$ protein was still more stable even in the elongation zone (Fig. 4, D and E) and showed strong resistance to $H_2O_2$ treatment (Fig. 5), suggesting that $H_2O_2$ induced the PLT2 protein degradation via the 212th cysteine. In contrast, the PLT1 protein distribution was limited around the RAM even after the RGF1 treatment (Fig. 1A), suggesting that the alterations of $H_2O_2$ levels by RGF1 are not coincident with the existence of the PLT1 protein and do not affect PLT1 protein distributions. The *plt1/plt2* mutant has severe root meristem defects (Galinha et al. 2007), suggesting that the *PLT1* and *PLT2* genes are primarily responsible for root meristem development because these 2 protein localizations overlapped around RAM (Fig. 1A). However, only PLT2 protein localization covered with the meristematic zone. Furthermore, the *PLT1* expression levels were significantly reduced in the *gPLT2-GFP/plt2* line after RGF1 treatment compared with the mock (Fig. 1G). In contrast, the *PLT1* expression was comparable in the *plt2* mutant even after RGF1 treatment relative to mock treatment, suggesting that the PLT2 accumulated by RGF1 transcriptionally reduced the *PLT1* expression. These results indicated that RGF1 transcriptionally decreased the PLT1 expression via PLT2 but did not enhance the PLT1 allocation through post-translational regulation via ROS. We thus found that the PLT2 has a distinct function under the RGF1 signaling pathway.

The distinct localizations of 2 ROS in individual developmental zones modulate root development as signaling molecules (Dunand et al. 2007; Tsukagoshi et al. 2010). ROS alterations affect various protein activities and stabilities. Various biological events happen in the specific root developmental zone. In this study, we focused only on PLT2. However, ROS are expected to posttranslationally regulate many proteins in specific developmental zones. Further cysteine proteome analysis in individual developmental zones may uncover target proteins that function via Cys through ROS required for root development in a specific root developmental zone. In summary, our current study provided insight into the post-translational regulation and modification of one of the pivotal root meristem regulators in a specific root developmental zone via the cysteine by ROS.

## Materials and methods

### Plant materials and growth conditions

All *Arabidopsis* mutants and transgenic lines used in this study were in the Columbia-0 (Col-0) background. The transfer (T)-DNA insertion line SAIL_846_C09 (*plt1*) was obtained from the Arabidopsis Biological Resource Center at Ohio State University. SALK_130119.20.25 (*plt2*) and SALK_127417.55.25 (*plt3*) were previously described (Galinha et al. 2007; Yamada et al. 2020). Seeds

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were surface-sterilized in 50% bleach coupled with 0.1% Tween 20 (Sigma) for 10 min, then rinsed 5 times with sterile water, followed by stratification at 4 °C for 2 d. The sterilized seeds were plated on half-strength MS medium (Murashige and Skoog salt mixture; Caisson Laboratories) containing 0.05% MES, 1% sucrose, and 1% agar (SIGMA), and that pH was adjusted to 5.7 with KOH. Seedlings were grown vertically in a growth incubator at 22 °C with 16 h light/8 h darkness cycles for 7 d. Then, seedlings were transferred to ½ MS agar plates containing either water (mock treatment), synthetic sulfated RGF1 peptide (Mission Biotech, Taiwan), or the indicated chemicals for 24 h for further experimentation.

## Plasmid constructions and transformations into plants

The genomic sequence of the *PLT2* gene, including a native promoter (2,410 bp) without a stop codon, was amplified using CloneAmp HiFi PCR Premix (Takara Bio, USA) from genomic DNA and then subcloned into the *pENTR/D/TOPO* vector (Invitrogen). The following primers were used for g*PLT2* amplification: pENTR gPLT2-F and pENTR gPLT2 Nostop-R. (All primer names and sequences in this study are listed in Supplementary Table S1.) The sequence of the *PLT2* gene in the *pENTR/D/TOPO* vector (gPLT2 pENTR) was confirmed by Sanger sequencing.

To generate $gPLT2^{C112S}$, $gPLT2^{C212S}$, and $gPLT2^{C112SC212S}$, site-directed mutagenesis of g*PLT2* was performed using the in-fusion HD cloning Kit (Takara Bio). The further reactions followed the in-fusion HD cloning Kit manual (Takara Bio). Two sets of forward and reverse primers for $gPLT2^{C112S}$ and $gPLT2^{C212S}$ and 3 sets of primers were constructed to introduce mutations at Cys112, Cys212, and Cys112Cys212.

$gPLT2^{C112S}$ fragment1: In-fusion gPLT2 forward new and PLT2 C112S-r

$gPLT2^{C112S}$ fragment2: PLT2 C112S-f and in-fusion gPLT2 reverse new

$gPLT2^{C212S}$ fragment1: In-fusion gPLT2 forward new and PLT2 C212S-r

$gPLT2^{C212S}$ fragment2: PLT2 C212S-f and in-fusion gPLT2 reverse new

$gPLT2^{C112SC212S}$ fragment1: In-fusion gPLT2 forward new and PLT2 C112S-r

$gPLT2^{C112SC212S}$ fragment2: PLT2 C112S-f and PLT2 C212S-r

$gPLT2^{C112SC212S}$ fragment3: PLT2 C212S-f and in-fusion gPLT2 reverse new.

To generate the linearized *pENTR* vector from the *pENTR* vector, inverse PCR was conducted using the following primers: inverse pENTR F and inverse pENTR R.

Two or 3 DNA fragments were fused to the linearized *pENTR* vector, confirming the sequences.

The clones of *gPLT2*, $gPLT2^{C112S}$, $gPLT2^{C212S}$, and $gPLT2^{C112SC212S}$ were recombined with the *pGWB450*[18] vector using Gateway LR Clonase II (Invitrogen). The constructs of *gPLT2-GFP*, $gPLT2^{C112S}$-*GFP*, $gPLT2^{C212S}$-*GFP*, and $gPLT2^{C112S\ C212S}$-*GFP* were transformed into the *plt2* mutant.

The *gPLT2-3xGFP* construct was made based on the *gPLT2-GFP* backbone by replacing the *GFP* with 3 copies of the *GFP* through in-fusion cloning. The constructs of *gPLT2-GFP*, $gPLT2^{C112S}$-GFP, $gPLT2^{C212S}$-*GFP*, $gPLT2^{C112S\ C212S}$-*GFP*, and *gPLT2-3xGFP* were transformed into *Agrobacterium tumefaciens* GV3101 strain and introduced into *Arabidopsis* (*plt2* background) using the floral dipping method (Clough and Bent 1998). Transgenic plants were selected on 50 μg/mL kanamycin plates, and the 2 independent T3 homozygous lines from each genotype were used in this study. The *gPLT1-YFP* and *gPLT2-YFP* lines have been previously generated (Galinha et al. 2007).

## Microscopic analysis

For detecting $O_2^-$, the seedlings were stained with 200 μM NBT dissolved in 20 mM phosphate buffer (pH 6.1) in the dark for 2 min and rinsed twice with distilled water (Yamada et al. 2020). Root images were taken using a 10× objective with a Zeiss AXIO Scope A1.

For detecting $H_2O_2$, the seedlings were stained with 25 μM BES-$H_2O_2$-Ac (Wako) for 10 min in the dark in a shaker and rinsed twice with distilled water. Root images were taken using a 20× objective with a Zeiss LSM 980.

The GFP and BES-$H_2O_2$-Ac or YFP signals were observed using a 20× objective with a Zeiss LSM 980 laser scanning confocal microscope. Excitation and detection parameters were set as follows: GFP and BES-$H_2O_2$-Ac, excitation at 488 nm and detection at 491 to 544 nm; YFP, excitation at 514 nm and detection at 517 to 544 nm; and PI staining, excitation at 561 nm and detection at 565 to 747 nm. The microscopic images were stitched and analyzed using the Fiji package of ImageJ (Thévenaz and Unser 2007; Schindelin et al. 2012). We define the meristematic zone as the region where the length of the root cortex cells is greater than their width and mark arrowheads.

## Measurement of the GFP signals at the specific sites in the developmental zones

We grew the seedlings of the *gPLT2-GFP/plt2* and $gPLT2^{C212S}$-*GFP/plt2* lines for 7 d on ½ MS agar plates. The single-layer images for GFP and PI were taken under the confocal laser scanning microscope (Zeiss LSM 980). The sites close to the apical meristem, near the elongation zone in the meristematic zone, and the elongation zone were identified as the apical meristem, the proximal side of the meristematic zone, and the elongation zone. The GFP intensities of 3 cells of 28 to 30 roots in the apical meristem (Fig. 4D, red arrowheads), the proximal side of the meristematic zone (Fig. 4D, yellow arrowheads), and the elongation zone (Fig. 4D, blue arrowheads) were measured with 3 biological replicates in the *gPLT2-GFP/plt2* and $gPLT2^{C212S}$-*GFP/plt2* lines using Fiji (Thévenaz and Unser 2007; Schindelin et al. 2012).

## Immunoblotting

Root tip tissues were harvested under a dissection microscope (Leica M205 FCA) using an ophthalmic scalpel (Feather) and homogenized in 1× SDS sample buffer containing 50 mM Tris-HCl (pH 6.8), 10% glycerol, 2% SDS, and 5% β-mercaptoethanol. The protein samples were incubated at 95 °C for 10 min. The samples were separated by 10% SDS-PAGE and transferred to 0.45 μm nitrocellulose membrane. The PLT2-GFP fusion protein abundance was detected by using anti-GFP (cat#632381, 1:3,000 dilution; Clontech) and anti-ACTIN (cat#MA1-744, 1:5,000 dilution; Thermo Fisher) as an internal control. The goat anti-mouse IgG-HRP conjugate (cat#1706516, 1:4,000 dilution; BIO-RAD) and the secondary antibody were used to detect anti-GFP and anti-ACTIN. The chemiluminescent signals were monitored by a ChemiDoc XRS+ System (Bio-Rad).

## Protein S-sulfenylation detection

Protein S-sulfenylation was measured using a biotin-switch method as described previously (Cao et al. 2024) with modifications. The 7-d-old XVE: PLT2-GFP seedlings were induced with 10 μm β-estradiol for 24 h followed by water or 1 mM $H_2O_2$ treatment

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for the indicated time, and then, the whole root tissues were harvested. Seedlings were protected from light during sample preparation until the pull-down step. Each step ensured the reaction was well mixed by vortex and rotation. The root extracts were incubated with 50 mM MMTS in the HENS (100 mM HEPES [pH 7.8], 1 mM EDTA, 0.1 mM neocuproine, 1% SDS) buffer at 50 °C for 1 h to block the free thiol groups. The supernatants were passed through Zeba spin desalting columns (Thermo Fisher) to remove excess MMTS and then incubated with 2 mM biotin-HPDP and 50 mM sodium arsenite at ambient temperature for 1 h to detect S-sulfenylation. The samples were passed through Zeba spin desalting columns to remove unreacted chemicals. An aliquot of protein was saved as input for western blot analysis; the rest was pulled down with NeutrAvidin agarose (Thermo Fisher) and eluted with 2× SDS sample buffer.

### RNA extraction and real-time qPCR analysis

Total RNA was isolated using the RNeasy Micro Kit (Qiagen). Briefly, the root tip tissues were homogenized in extraction buffer and processed following the manufacturer's protocol. The concentration of total RNA was measured by a Qubit (Invitrogen) instrument. Reverse transcription was performed using SuperScript IV Reverse Transcriptase (Invitrogen). The SYBR Green Supermix (BIO-RAD) was used with a CFX Connect Real-Time PCR Detection System (BIO-RAD) for qPCR. Relative transcript levels were determined by normalizing with *PP2AA3* (At1g13320). Three biological replicates and technical replicates were used for each experiment.

### RNA-seq library preparation and data processing

RNA quality was examined using a 2100 Bioanalyzer (Agilent). The RNA integrity number was above 8.0 in all samples. RNA-seq libraries were generated from 100 ng total RNAs using Universal RNA-Seq with NuQuant library preparation kit (Tecan Trading AG) following the manufacturer's protocols. The libraries for 5 to 6 biological replicates of mock- and RGF1-treated root samples were sequenced on an Illumina NextSeq 2000 (100 base paired-end reads). The bulk RNA-sequencing reads were aligned to the *Arabidopsis* reference genome TAIR10 using STAR (Dobin et al. 2013). The mRNA read counts of genes were computed using featureCounts (Liao et al. 2014). Genes expressed (read counts > 0) are included for analysis. Normalization for sequencing depth and differential gene analysis was performed using DESeq2 (Love et al. 2014). Genes with FDR < 5% were considered differentially expressed. GO analysis was performed using the Bioconductor package clusterProfiler. All DEGs for bulk RNA sequencing can be found in Supplementary Data S1.

### Statistical analysis and reproducibility

Experiments were independently repeated 3 times with similar results. No power analysis was done to estimate the sample size. All statistical analyses were performed using R version 4.3.2 (http://www.r-project.org/). For 2-sample comparison, data were analyzed with a 2-tailed *t*-test. As an omnibus test, a Welch's analysis of variance (ANOVA) or an aligned rank transform ANOVA (ART) was performed, dependent on the data normality. As a post hoc test, ART-Contrast test was applied to nonnormal data, and the Games–Howell test or Tukey's HSD test was performed on normal data, depending on the data homoscedasticity. In the boxplots, the circles represent each individual data point; the horizontal line within boxes represents the median; the upper and lower hinges, respectively, indicate the 75th and 25th percentiles (interquartile range [IQR]); and the whiskers show the 1.5 extension of the IQR. Data points out of the whiskers were considered outliers.

### Accession numbers

Sequence data from this article can be found in the GenBank/EMBL data libraries under accession number GSE244175.

## Acknowledgments

We thank Miranda Loney and Hironaka Tsukagoshi for their comments on the manuscript; Yi-Han Weng for taking the confocal images; Renze Heidstra for the *gPLT2-YFP*, *gPLT1-YFP*, and *plt3* seeds; the High Throughput Sequencing Core in the Biodiversity Research Center at Academia Sinica for sequencing the Illumina libraries; the Confocal Microscope Core Facility in the Biotechnology Center in Southern Taiwan at Academia Sinica for maintaining the confocal laser scanning microscope; the DNA Sequencing Core Facility (Core Facility and Innovative Instrument Project, AS-CFII-113-A12) for the DNA Sequencing support; and NGS High Throughput Sequencing Core Facility for maintaining the server at Academia Sinica.

## Author contributions

Y.-C.H. and M.Y. conceptualized the study; Y.-C.H., S.-Y.S., and M.Y. performed all experiments; M.-R.Y. and J.-K.L. performed the computational and statistical analyses; all the authors wrote the paper.

## Supplementary data

The following materials are available in the online version of this article.

**Supplementary Figure S1.** Global sequence alignment of PLT1 and PLT2.

**Supplementary Figure S2.** RGF1 similarly modulates the ROS level in each PLT2 reporter line.

**Supplementary Figure S3.** The second independent lines of each PLT2 construct show a similar expression pattern and response to RGF1 treatment.

**Supplementary Figure S4.** The protein localization of PLT2-1xGFP and PLT2-3xGFP after RGF1 treatment.

**Supplementary Figure S5.** ROS affect the RGF1-dependent PLT2 stability via cysteine residues.

**Supplementary Figure S6.** Schema depicting the role of RGF1-ROS in regulating root meristem development via the 212th Cys of PLT2.

**Supplementary Table S1.** Primer list.

**Supplementary Data S1.** The normalized count values and DEGs list (DEGs_norm-count).

**Supplementary Data S2.** GO terms.

## Funding

This work was funded by the Academia Sinica Taiwan (AS-CDA-111-L05); the National Science and Technology Council Taiwan (112-2311-B-001-021-MY3, 111-2311-B-001-028, and 109-2313-B-001-006-MY3); and the Agricultural Biotechnology Research Center, Academia Sinica, Taiwan to M.Y.

*Conflict of interest statement*. None declared.

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## Data availability

RNA-seq data have been deposited at GEO: GSE244175 and are publicly available as of publication.

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